ARTIFICIAL INTELLIGENCE

CSE 3201

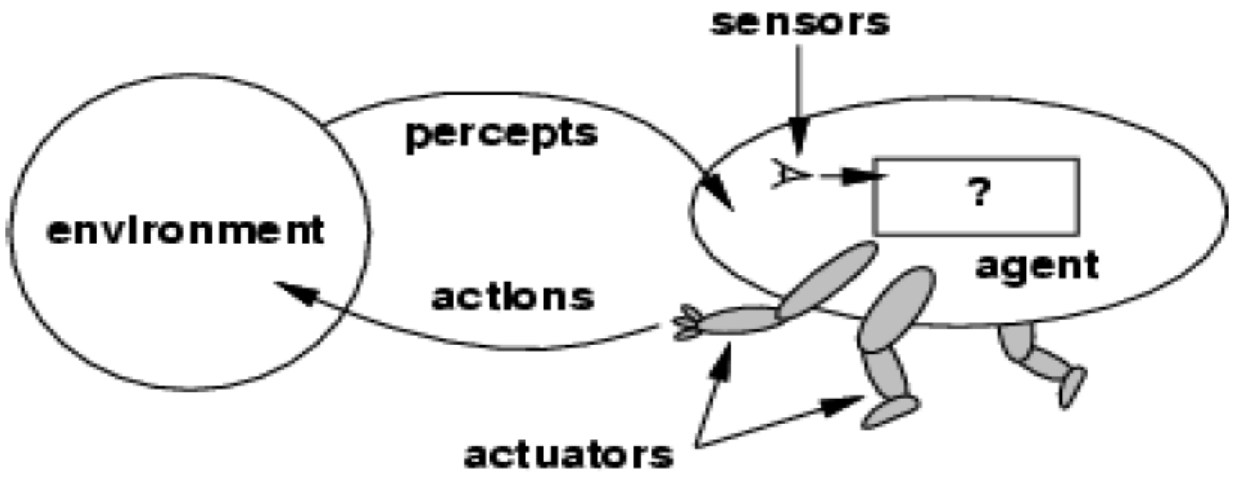
Intelligent Agents

Outline

- Agents and Environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types

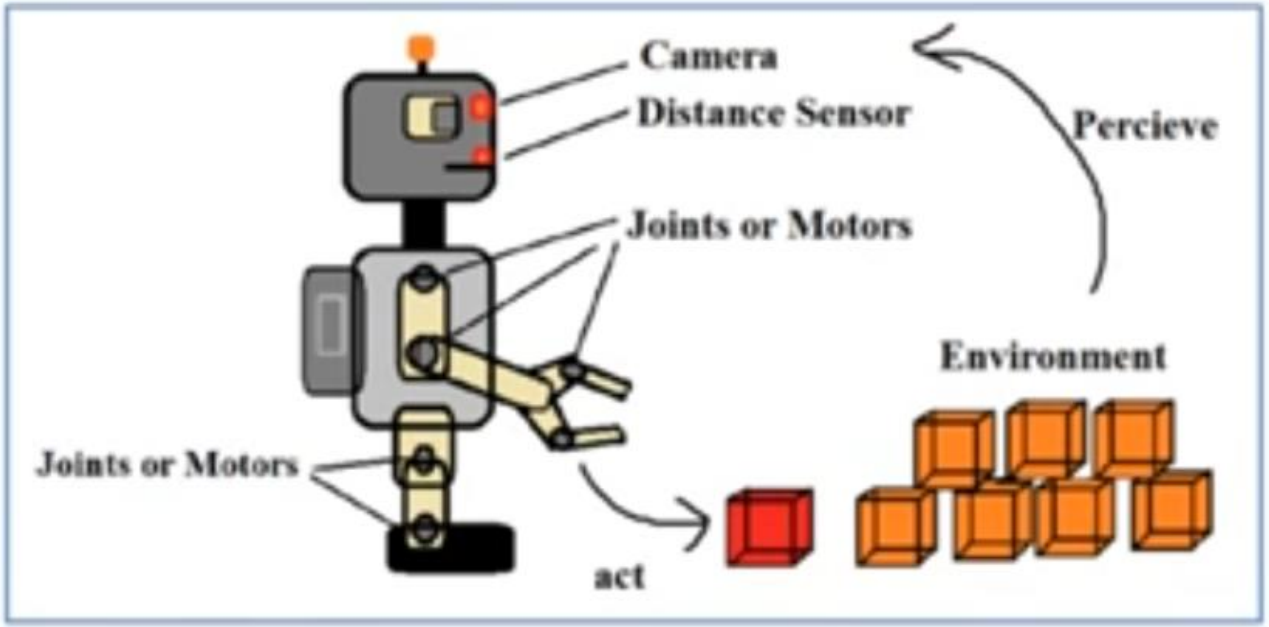
Agents

- An agent is anything that can be viewed as
  - perceiving its environment through sensors and
  - acting upon that environment through actuators
  - Assumption: Every agent can perceive its own actions (but not always the effects)



Agents

- Agent carries 3 key tasks continuously in a loop or cycle. They are Sense, Think and Act. It is called "sense-think-act" cycle or "perception-action" cycle.
- Example: Robotics Agent.



Agents

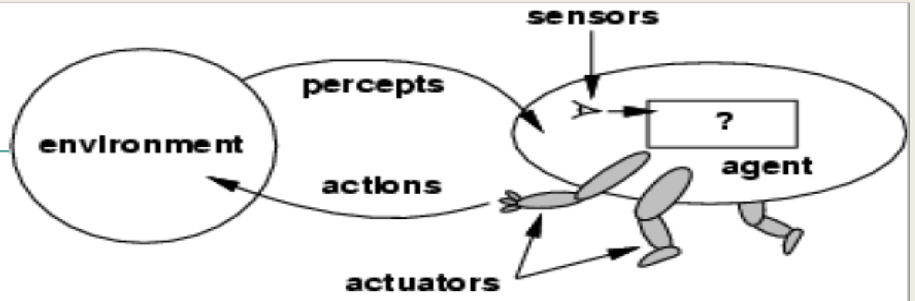
- Human agent:
  - eyes, ears, and other organs for sensors;
  - hands,legs, mouth, and other body parts for actuators
- Robotic agent:
  - cameras and infrared range finders for sensors;
  - various motors for actuators

Intelligent Agents

Four main aspects that need to be taken into consideration when designing an intelligent agent.

- Percepts: This is the information that the agent receives
- Actions: This is what the agent needs to do or can do to achieve its objectives.
- Goals: This is the factor that the agent is trying to achieve
- Environment: The final aspect is the environment in which the agent will be working in. The environment in which the agent performs is probably the most important aspect that needs to be considered as this affects the outcome of the percepts, actions and goals.

Agents and Environment



- Percept: agent's perceptual input at any given instant
- Percept sequence: complete history of everything the agent has ever perceived
- An agent's choice of action at any given instant can depend on the entire percept sequence observed to date

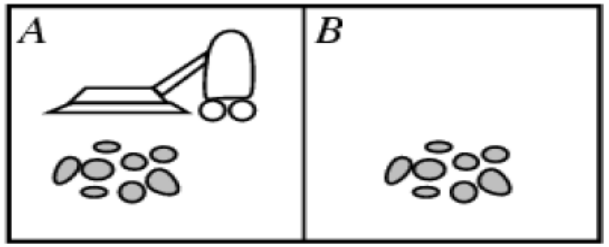
- An agent's behavior is described by the agent function which maps from percept histories to actions:

$$[f: P^* \rightarrow A]$$

Environment -> sensors -> **agent function** -> actuators -> environment.

- We can imagine tabulating the agent function that describes any given agent (External characterization)
- Internally, the agent function will be implemented by an agent program which runs on the physical architecture to produce f
- agent = architecture + program

Vacuum-Cleaner World



- Two locations: A and B
- Percepts: location and contents, e.g., [A,Dirty]
- Actions: *Left*, *Right*, *Suck*, *NoOp*

Percept sequence	Actions
[A,Clean]	Right
[A, Dirty]	Suck
[B,Clean]	Left
[B,Dirty]	Suck
[A,Clean],[A,Clean]	Right(Act on a single perception)
[A,Clean],[A,Dirty]	Suck
…	…
[A,Clean],[A.Clean],[A,Clean]	Right
[A,Clean],[A,Clean],[A, Dirty]	Suck

(Act on a sequence of perception)

One simple function is :

if the current square is dirty then suck, otherwise move to the other square

Rational Agent

- An agent should strive to "do the right thing", based on what it can perceive and the actions it can perform.
- The right action is the one that will cause the agent to be most successful
- Performance measure: An objective criterion for success of an agent's behavior
- E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.
- As a general rule, it is better to design performance measures according to what one actually wants in the environment. Rather than according to how one thinks the agent should behave (amount of dirt cleaned vs a clean floor)
- A more suitable measure would reward the agent for having a clean floor

Rational Agent

- What is rational at any given time depends on four things
  - The performance measure that defines the criterion of success
  - The agent's prior knowledge of the environment
  - The actions that the agent can perform
  - The agent's percept sequence to date

  - Rational Agent: For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Vacuum Cleaner Agent

- Let's assume the following
  - The performance measure awards one point for each clean square at each time step, over a lifetime of 1000 time steps
  - The geography of the environment is known a priori but the dirt distribution and the initial location of the agent are not. Clean squares stay clean and the sucking cleans the current square. The Left and Right actions move the agent left and right except when this would take the agent outside the environment, in which case the agent remains where it is
  - The only available actions are Left, Right, Suck and NoOp
  - The agent correctly perceives its location and whether that location contains dirt
- Under these circumstances the agent is rational:its expected performance is at least as high as any other agent's

Specifying the task environment (PEAS)

- PEAS:
  - **P**erformance measure,
  - **E**nvironment,
  - **A**ctuators,
  - **S**ensors
- In designing an agent, the first step must always be to specify the task environment (PEAS) as fully as possible

PEAS for an automated taxi driver

- Performance measure: Safe, fast, legal, comfortable trip, maximize profits
- Environment: Roads, other traffic, pedestrians, customers
- Actuators: Steering wheel, accelerator, brake, signal, horn
- Sensors: Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

PEAS for a medical diagnosis system

- Performance measure: Healthy patient, minimize costs, lawsuits
- Environment: Patient, hospital, staff
- Actuators: Screen display (questions, tests, diagnoses, treatments, referrals)
- Sensors: Keyboard (entry of symptoms, findings, patient's answers)

PEAS for Interactive English tutor

- Performance measure: Maximize student's score on test
- Environment: Set of students
- Actuators: Screen display (exercises, suggestions, corrections)
- Sensors: Keyboard

PEAS for Mushroom-Picking Robot

Performance Measure	Environment	Actuators	Sensors
Percentage of good mushrooms in correct bins	Conveyor belt with mushrooms, bins	Jointed arm and hand	camera, joint angle sensors



Omniscience

- Usually when we think of omniscience, we're thinking of an attribute of God. But in artificial intelligence, omniscience has a different definition.
- An omniscient agent knows the *actual effects of its actions.*
- In other words, any AI program that knows the exactly what will happen for all its possible actions is omniscient. By this definition, a good Tic-Tac-Toe AI is omniscient because it always knows the outcome.
- Rationality is not the same as perfection. Rationality maximizes *Expected* performance , while perfection maximizes *Actual* performance.

Environment types

Fully observable vs. partially observable:

- An environment is fully observable if an agent's sensors give it access to the complete state of the environment at each point in time.
- Fully observable environments are convenient, because the agent need not maintain any internal state to keep track of the world
- An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data
- Examples: vacuum cleaner with local dirt sensor, taxi driver

Environment types

Fully vs Partially Observable (Cont…)

- Consider the example of Chess where each player has access to the complete board information. Every decision is made considering the state of the board at that time and the possible moves by the other player. This is a ‘fully observable’ environment.
- Contrast this with Poker where players cannot anticipate the opponent’s game as they do not have access to the opponent’s cards. Such an environment is ‘Partially Observable’.

Environment types

Deterministic vs Stochastic

- In a Deterministic environment, the outcome is certain and can be determined based on a specific state. However, in a Stochastic environment, the next state cannot be predicted with certainty. Chess can be considered as having a Deterministic environment. The autonomous vehicle, however, has a Stochastic environment. Here, the decision taken by the autonomous vehicle is based on the probability of action of other vehicles.
- In a deterministic environment, any action that is taken uniquely determines its outcome. In chess, for example, moving a pawn from A2 to A3 will always work. There is no uncertainty.
- In a stochastic environment, there is always some level of randomness. Any games that involve dice are good examples - you can never be certain that a specific number will be rolled.

Environment Types

Episodic vs Sequential

- An episode = Agent's single pair of perception & action.
- The quality of the agent's action doesn't depend on other episodes. That is, the next episode does not depend on the actions taken in previous episodes.
- Episodic environment is simpler, the agent doesn't need to think ahead.
- In case of sequential, current action may affect all future decisions.

  - Exp: Taxi Driver and Chess.

Environment Types

Static vs Dynamic

- A dynamic environment always changes over time.
- A Dynamic environment might change while the agent is processing a response. However, in a Static environment, the environment does not change during the course of decision making.

  - Ex: the number of people always changes in a street.

- In Chess, the environment remains unchanged (Static) during the period a player is attempting for the next move.

Environment types

Discrete vs. continuous:

- A limited number of distinct, clearly defined states, percepts and actions.
- Examples: Chess has finite number of discrete states, and has discrete set of percepts and actions. Taxi driving has continuous states, and actions

Single agent vs. multiagent:

- An agent operating by itself in an environment is single agent
- Examples: Crossword is a single agent while chess is two-agents

*See book Russel and Norvig (3rd Edition): figure 2.6

Agent Types

- Four basic types in order of increasing generality:
  - Simple reflex agents
  - Model-based reflex agents
  - Goal-based agents
  - Utility-based agents

Simple reflex agents

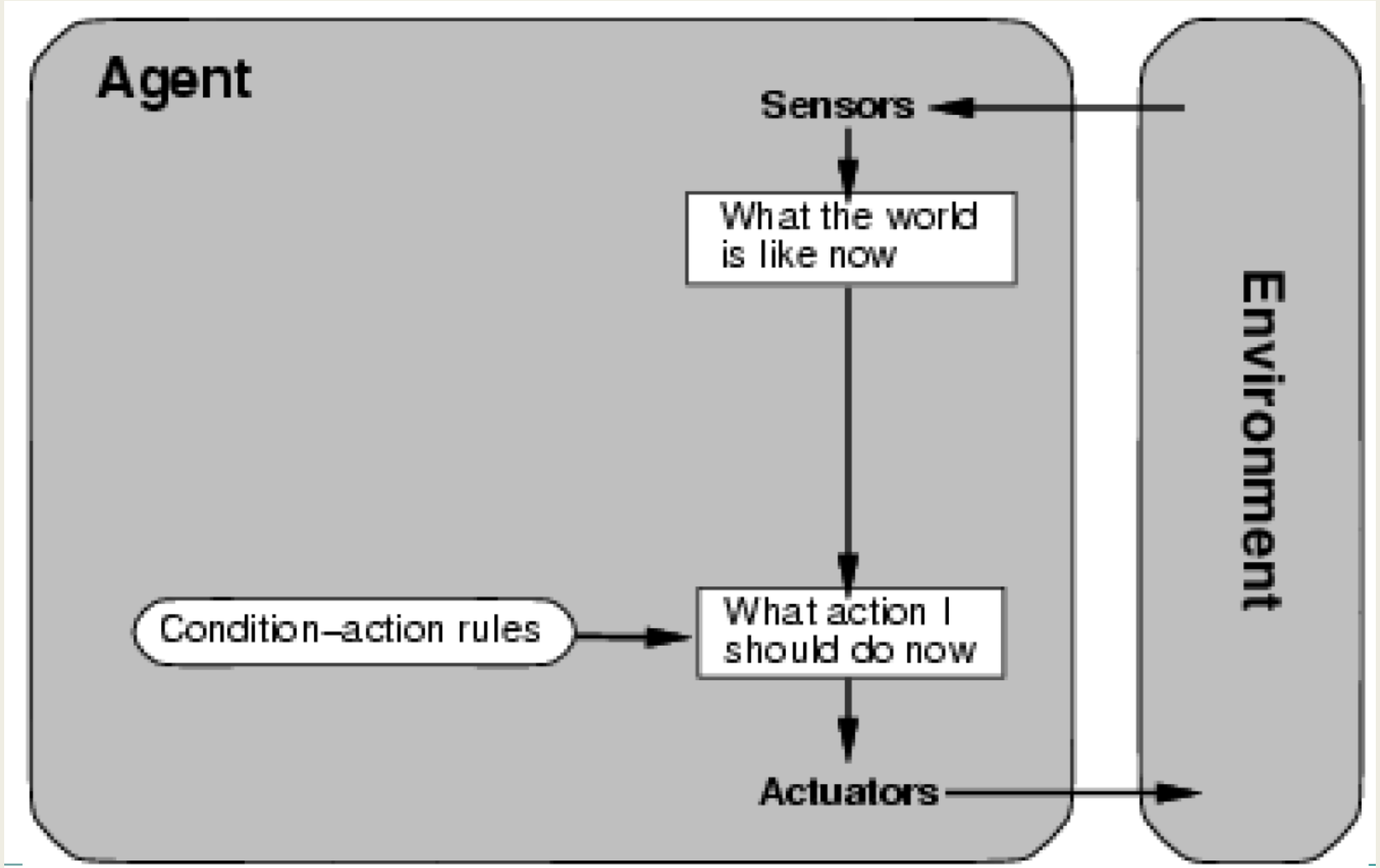
- Select actions on the basis of the **current percept** ignoring the rest of the percept history

- Example: simple reflex vacuum cleaner agent

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

- **Condition-action-rule**
- Example: **if** *car-in-front-is-breaking* **then** *initiate-breaking*

Simple reflex agents



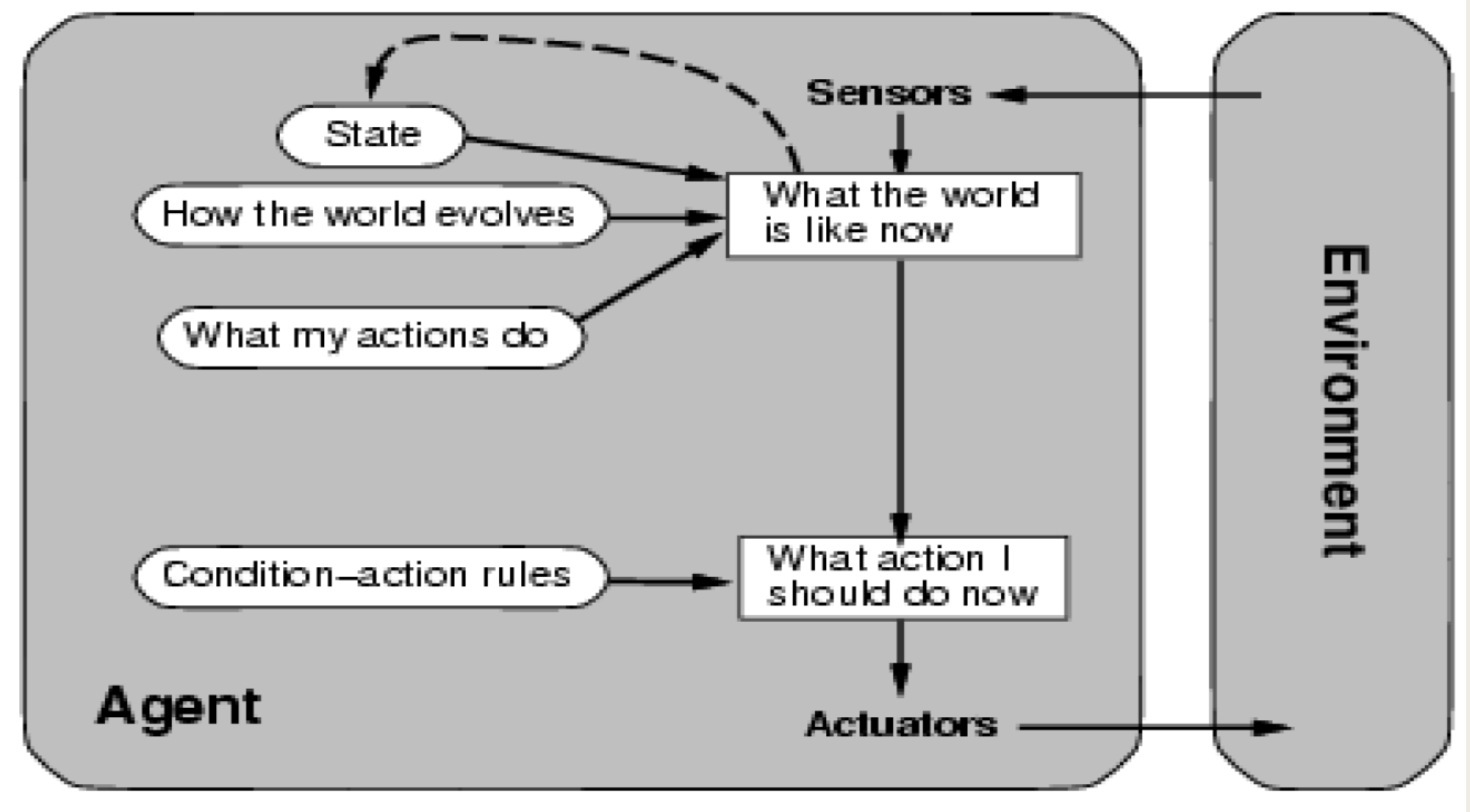
Simple reflex agents

- Have no Memory.
- Simple-reflex agents are simple, but they turn out to be of very limited intelligence.
- The agent will work only if the correct decision can be made on the basis of the current percept – that is only if the environment is fully observable.

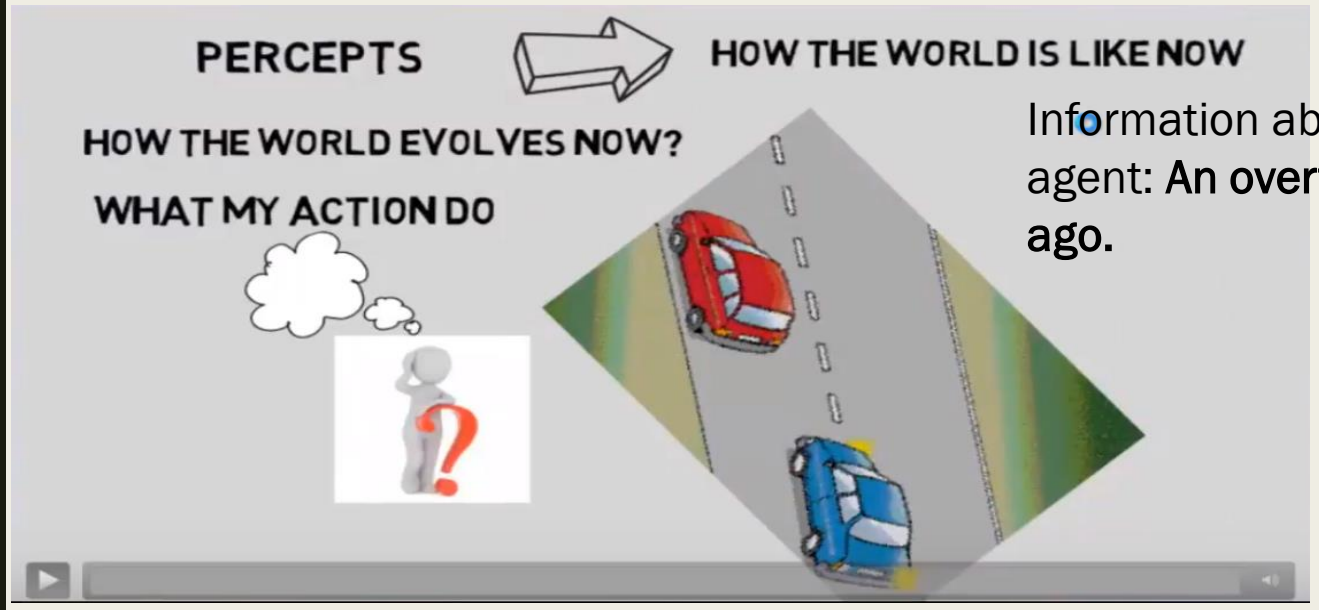
Model-based reflex agents

- The agent should keep track of the part of the world it can't see now
- The agent should maintain some sort of internal state that depends on the percept history and reflects at least some of the unobserved aspects of the current state
- Updating the internal state information as time goes by requires two kinds of knowledge to be encoded in the agent program
  - Information about how the world evolves independently of the agent
  - Information about how the agent's own actions affects the world

Model-based reflex agents



Model-based reflex agents



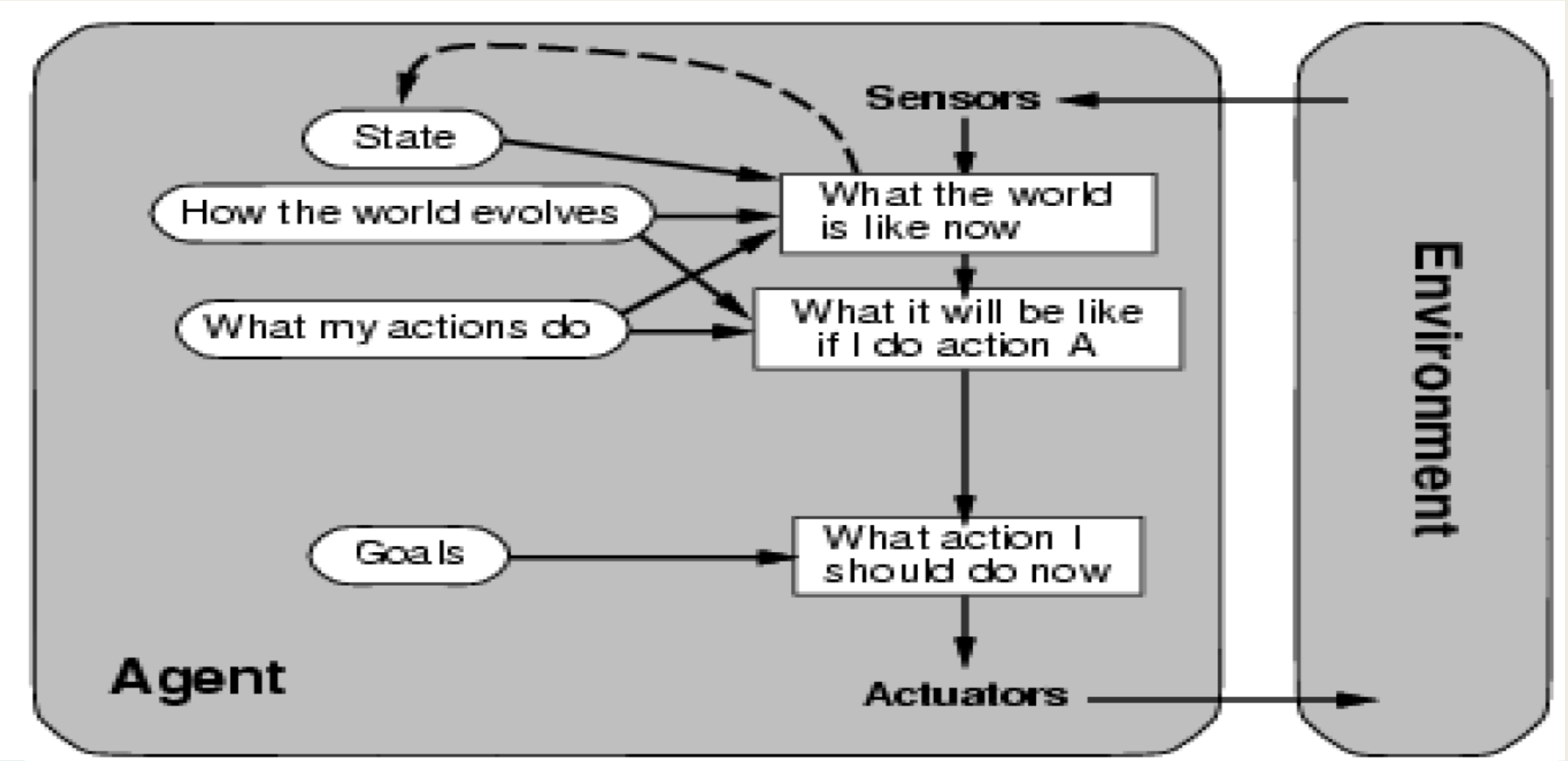
Information about how the world evolves independently of the agent: **An overtaking car generally be closer than it was a moment ago.**

So, in a nut shell we can say that model based reflex agent maintains the internal state to take the decision what action should take upon the environment. These agents are extension of condition action rule based reflexive agent.

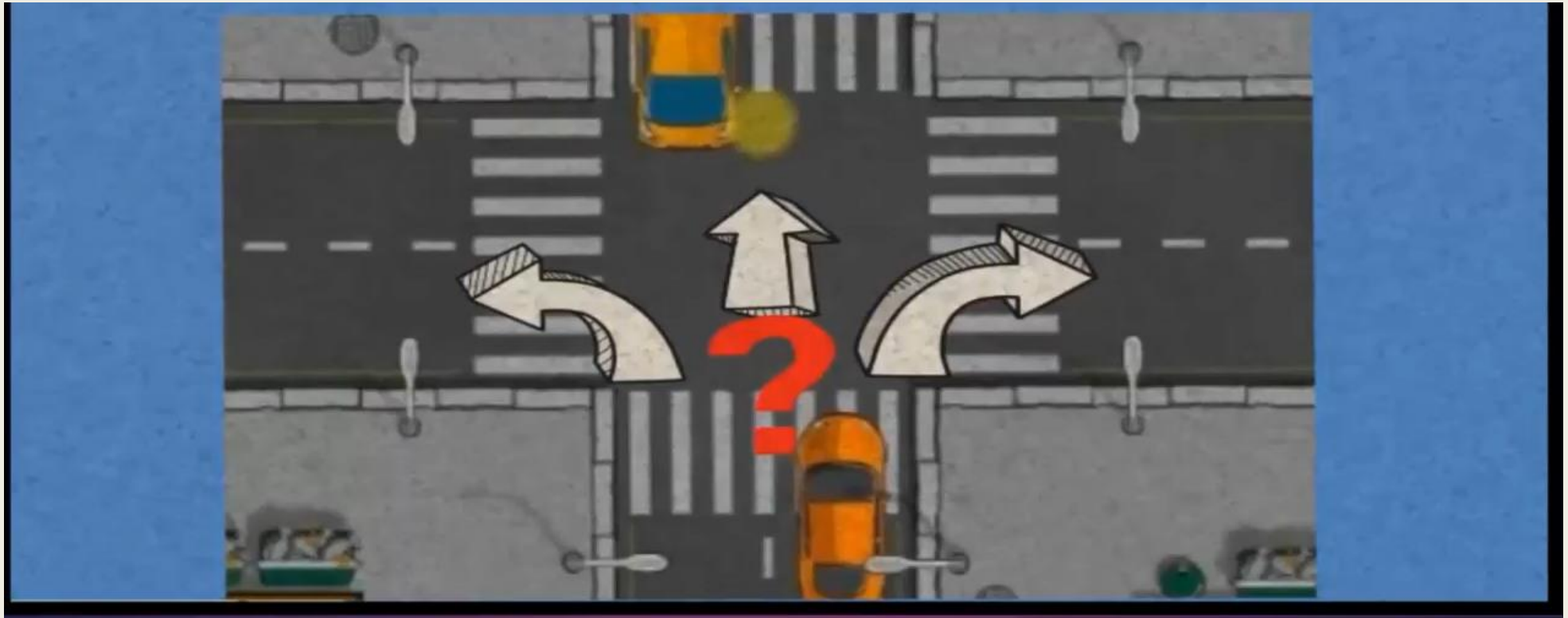
Goal-based agents

- Goal-based agents: Goal information needed to make decision.
- The agent program can combine this with information about the results of possible actions in order to choose actions that achieve the goal
- Involves consideration of the future, "what will happen if I do...?"
- Usually requires search and planning
- Knowing something about the current state of the environment is not always enough to decide what to do. In this situation goal will help you to take decision.

Goal-based agents



Goal-based agents



For example, at a road junction, the taxi can turn left, turn right, or go straight on.

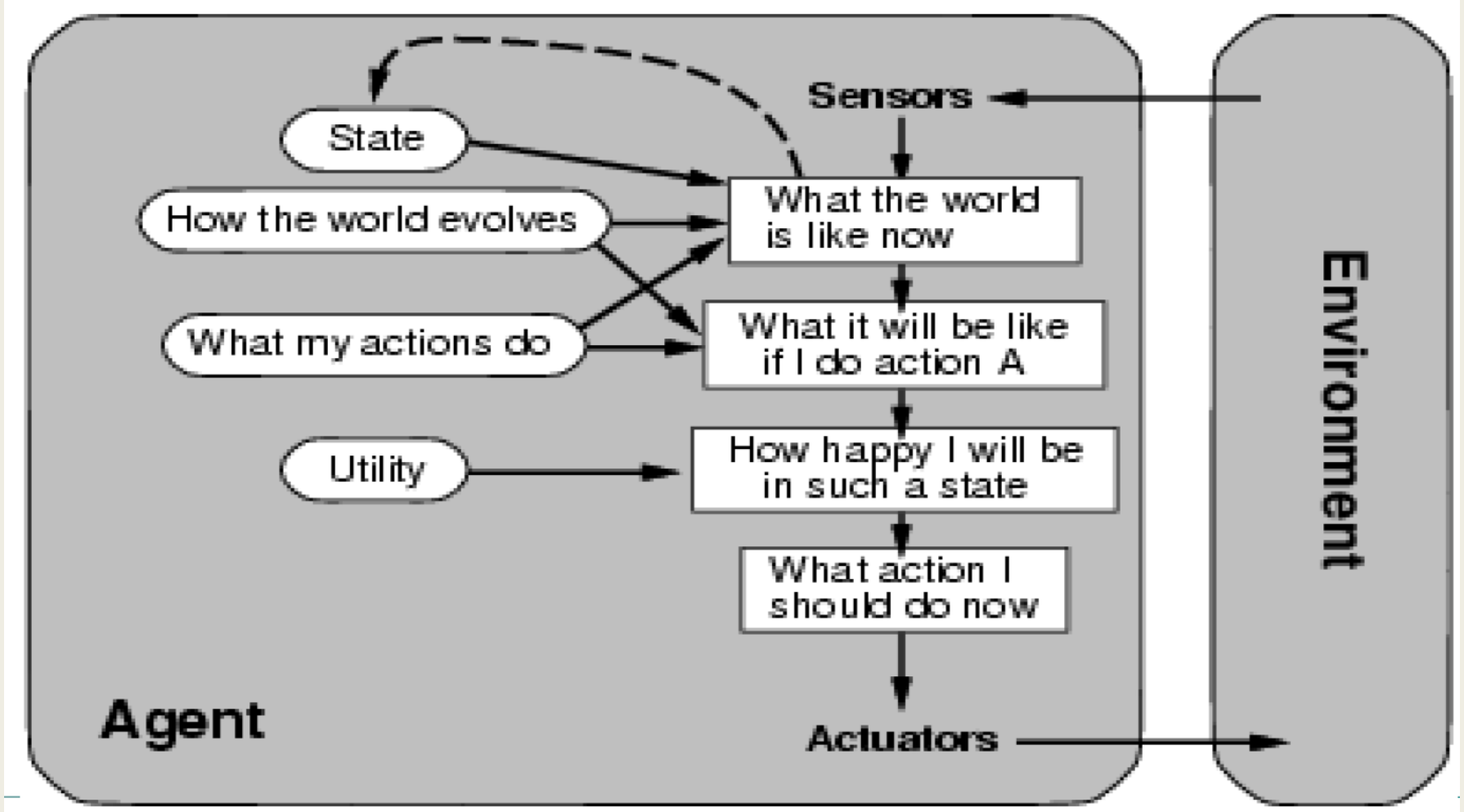
The correct decision depends on where the taxi is trying to get to.

In other words, as well as a current state description, the agent needs some sort of **goal** information that describes situations that are desirable.

Utility-based agents

- Goals alone are not really enough to generate high quality behavior in most environments – they just provide a binary distinction between happy and unhappy states
- A more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent if they could be achieved
- Happy – Utility (the quality of being useful)
- A utility function maps a state onto a real number which describes the associated degree of happiness

Utility-based agents



Goal-based vs Utility-based agent

- Well Goal Based agents will use information about what they know and their current state to see if they have accomplished what they wanted to or not however that alone is not enough, you also need a utility based agent to the performance of accomplishing the goal.
- For example, if you have an autonomous car that you want to take you somewhere, once it gets to a stop and it has the ability to turn in 3 different directions, using the goal-based agent it can know which one it should take to get to it's destination. Using the utility-based agent it can measure which direction might be better to get to the destination. Maybe one is faster but dangerous and the other is slower but safer.